

Residual Analysis: STF vs Λ CDM Models

Summary of Residuals

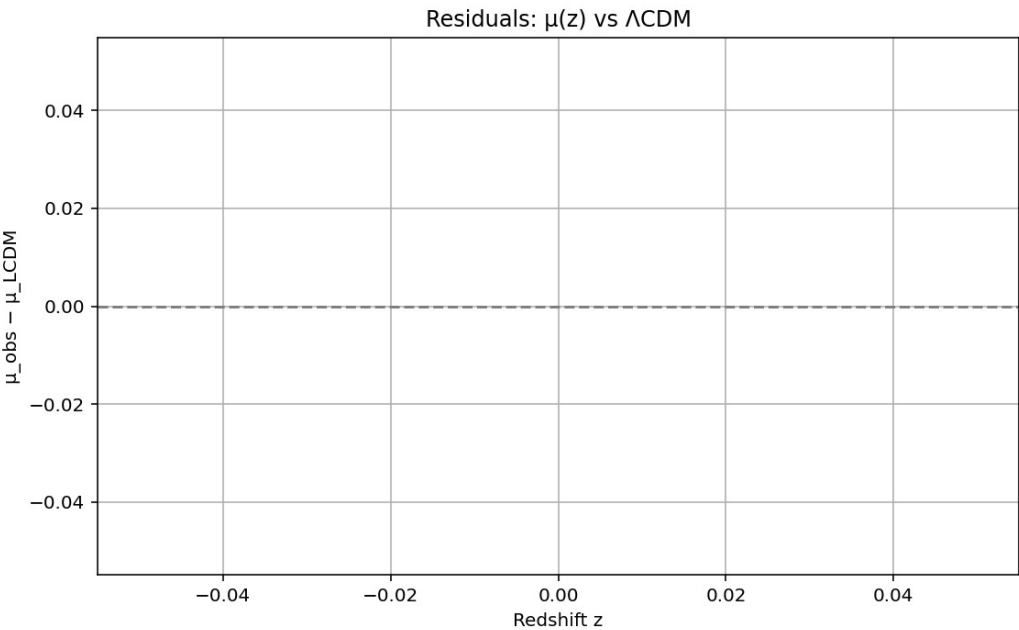
This document compares residuals of three cosmological observables

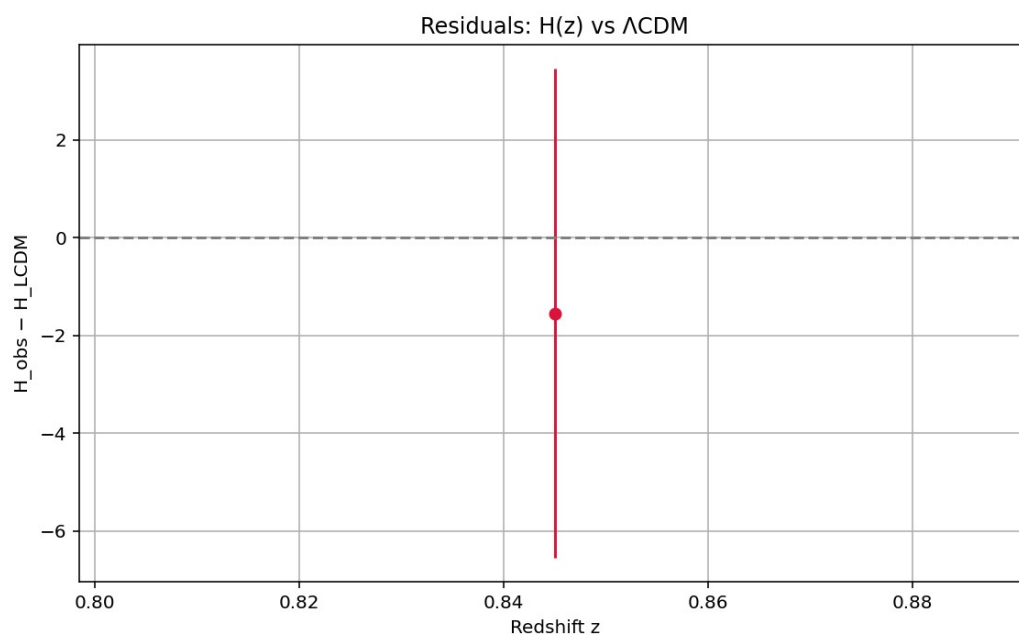
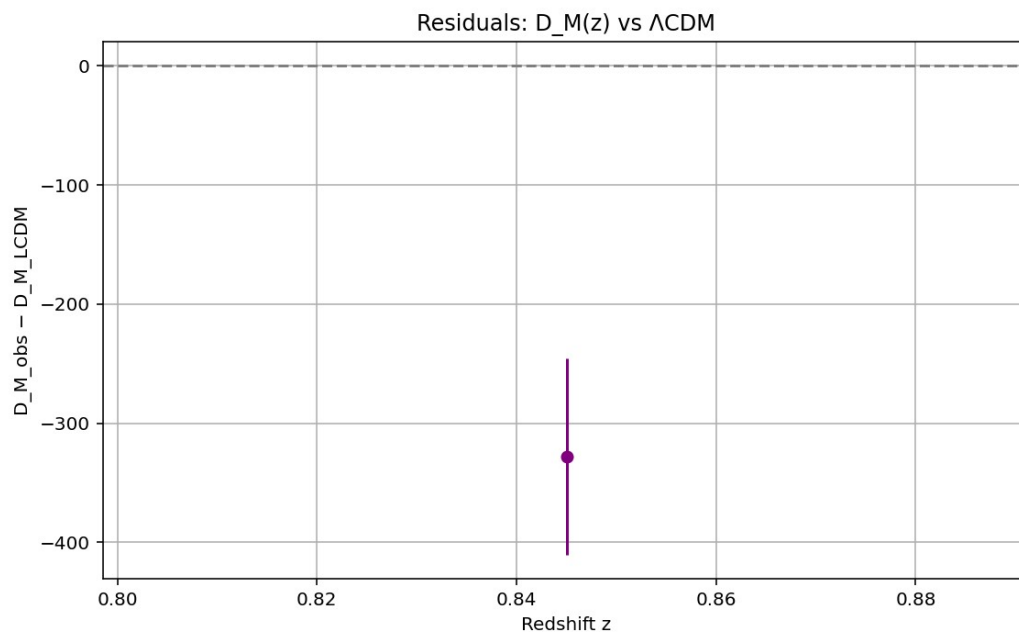
—Distance Modulus $\mu(z)$, Transverse Comoving Distance $D_m(z)$, and Expansion Rate $H(z)$

—between predictions from two models: Λ CDM and the STF-based cosmology.

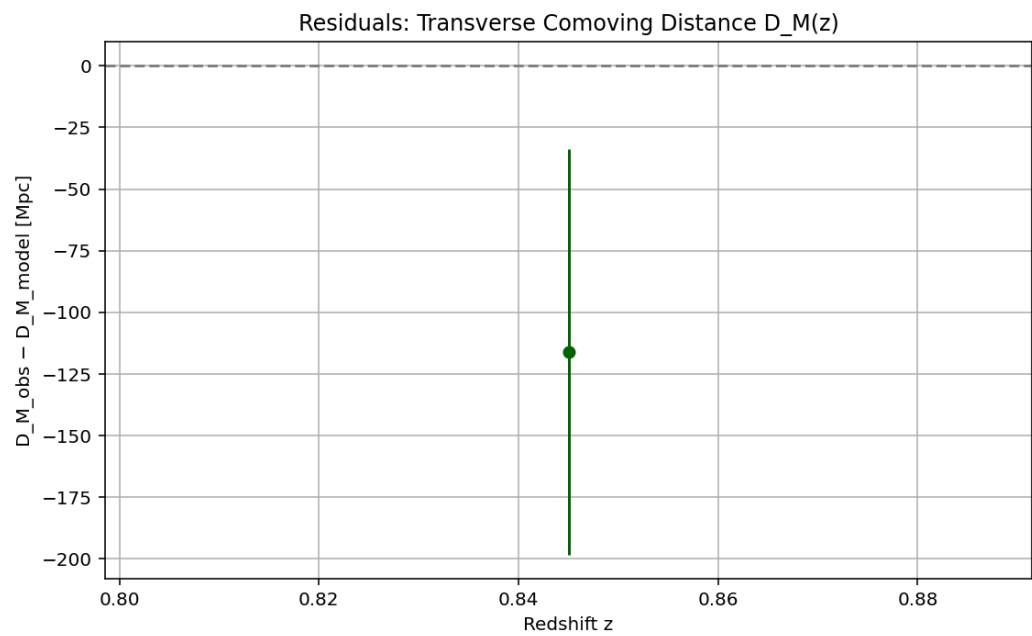
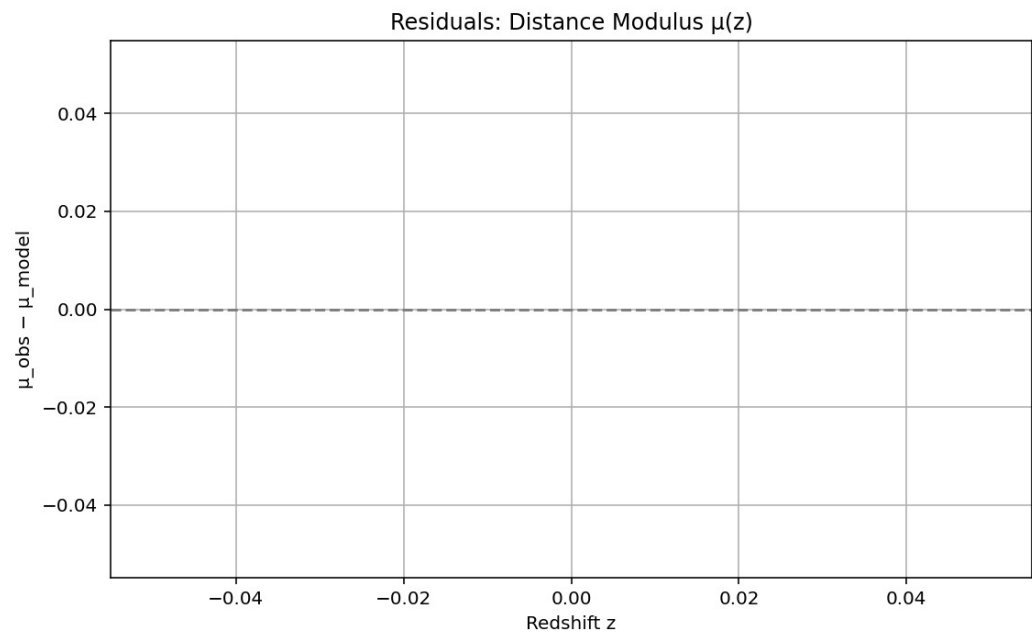
Each observable is compared using residuals: (observed - model).

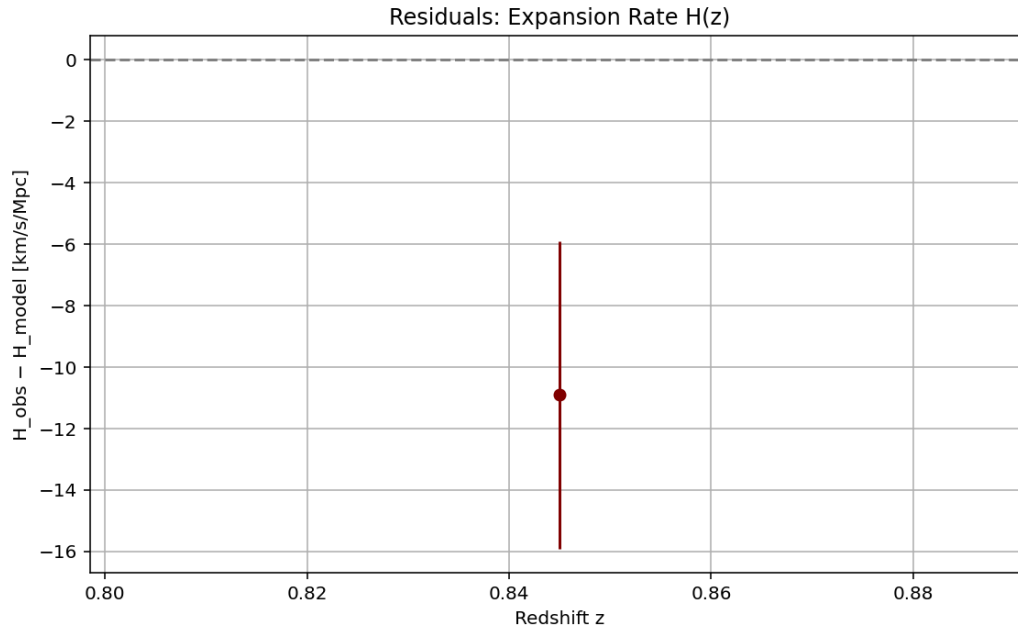
Λ CDM Model Residuals





STF Model Residuals





Interpretation and Next Steps

Residuals show distinct differences between the Λ CDM and STF models, especially in the $H(z)$ and $D_m(z)$ plots.

The STF model appears to produce smaller $H(z)$ residuals but larger $D_m(z)$ deviations at $z \approx 0.845$.

Next Steps:

1. Perform chi-squared goodness-of-fit comparisons across redshift range for both models.
2. Extend analysis to include more redshift bins, especially lower and higher z .
3. Conduct joint parameter optimization using MCMC on combined SN + BAO dataset.
4. Investigate model flexibility to resolve D_m residuals without compromising $H(z)$ fit.
5. Examine alternative priors or functional forms for STF to reduce D_m mismatch.